

Report R2 — Unitary sweep and fragmentation scaling

Krylov sector analysis of quantum cellular automata (Tier 1a/1c)

QCA Sector Analysis project (Tier 1)

2026-07-12 (engine_version 1a.1)

Abstract

We sweep all sixteen unitary quantum elementary cellular automata over chain lengths $N = 6 \dots 18$, both boundary conventions, and classify each by the scaling of its Krylov-sector structure. Sector sizes are exact; the growth of the number of sectors and of the largest sector $D_{\max}$ is fit by BIC model selection ($\ln y = c [+ \alpha \ln N] [+ \kappa N]$). Seven rules are effectively ergodic under `obc0` and nine under `pbk`; that verdict originally rested on $N = 6$ alone, and §4 re-establishes it over $N = 6 \dots 13$ with the full sector decomposition, where it turns out to be exactly N -independent and derivable from the rule encoding. The remaining rules fall into three fragmentation families: exponential (156/198, 108, 201), a domain-wall family with a linearly growing sector count but an extensive $D_{\max} = 2^{N-1}$ (60/102, `obc0` only), and 150/105 with central-binomial sectors. We highlight two boundary-sensitive phenomena and one broken symmetry.

1 Method

Each unit (rule, N , bc) is computed by the exact Tier-1a engine (Report R1) and cached as one JSON line in `results/`. Unitary sectors are the weakly connected components of the one-cycle transition graph (extracted by `union-find`; see R1). A rule is flagged *effectively ergodic at N* when one sector exceeds 90% of the 2^N states; the sweep then stops increasing N for that rule/bc. Growth classes are assigned to the series $y(N) \in \{\#\text{sectors}, D_{\max}\}$ by BIC selection among constant / power-law / exponential models, with the $\alpha \ln N$ prefactor always carried when reporting the exponential rate κ (so that $\sqrt{N}$ -type binomial prefactors do not bias κ).

2 Results

Table 1 lists every computed rule/bc with its sector count and $D_{\max}$ at the largest N reached and the fitted growth classes. Figures 1–3 show the scaling.

3 A caution on reading bases off these fits: rule 156

The growth-base map invites reading a constant off each point, and 156/198 is the case where that fails. Its $D_{\max}$ series (`obc0`) is

$$6, 9, 12, 16, 20, 27, 36, 48, 64, 81, 108, 144, 192 \quad (N = 6 \dots 18),$$

and it satisfies *both* $a(N+4) = 3a(N)$ and $a(N+5) = 4a(N)$ with exactly the same two exceptions ($N = 6, 10$). The implied bases are $3^{1/4} = 1.31607$ and $4^{1/5} = 1.31951$ — they differ by 0.26%, and thirteen terms cannot choose between them. No fit should be allowed to.

W	tuple	bc	$N_{\max}$	#sec	$D_{\max}$	#sec growth	$D_{\max}$ growth (base)
150	IVVI	obc0	18	10	92378	polynomial	exponential (2.011)
60	IIVV	obc0	18	19	131072	polynomial	exponential (2.000)
102	IVIV	obc0	18	19	131072	polynomial	exponential (2.000)
105	VIIV	obc0	18	10	92378	polynomial	exponential (1.973)
201	VIII	obc0	18	17991	6765	exponential	exponential (1.618)
108	IIIV	obc0	18	31572	2584	exponential	exponential (1.617)
156	IIVI	obc0	18	6765	192	exponential	exponential (1.281)
198	IVII	obc0	18	6765	192	exponential	exponential (1.281)
204	IIII	obc0	18	262144	1	exponential	constant
51	VVVV	obc0 ^{†6}	–	–	–	ergodic	ergodic
54	IVVV	obc0 ^{†6}	–	–	–	ergodic	ergodic
57	VIVV	obc0 ^{†6}	–	–	–	ergodic	ergodic
99	VVIV	obc0 ^{†6}	–	–	–	ergodic	ergodic
147	VVVI	obc0 ^{†6}	–	–	–	ergodic	ergodic
153	VIVI	obc0 ^{†6}	–	–	–	ergodic	ergodic
195	VVII	obc0 ^{†6}	–	–	–	ergodic	ergodic
105	VIIV	pbc	18	9	97240	constant	exponential (1.998)
150	IVVI	pbc	18	12	87516	polynomial	exponential (1.888)
108	IIIV	pbc	18	24914	5778	exponential	exponential (1.619)
201	VIII	pbc	18	24914	5778	exponential	exponential (1.619)
156	IIVI	pbc	18	5779	144	exponential	exponential (1.293)
198	IVII	pbc	18	5779	144	exponential	exponential (1.293)
204	IIII	pbc	18	262144	1	exponential	constant
51	VVVV	pbc ^{†6}	–	–	–	ergodic	ergodic
54	IVVV	pbc ^{†6}	–	–	–	ergodic	ergodic
57	VIVV	pbc ^{†6}	–	–	–	ergodic	ergodic
60	IIVV	pbc ^{†6}	–	–	–	ergodic	ergodic
99	VVIV	pbc ^{†6}	–	–	–	ergodic	ergodic
102	IVIV	pbc ^{†6}	–	–	–	ergodic	ergodic
147	VVVI	pbc ^{†6}	–	–	–	ergodic	ergodic
153	VIVI	pbc ^{†6}	–	–	–	ergodic	ergodic
195	VVII	pbc ^{†6}	–	–	–	ergodic	ergodic

Table 1: Per-rule fragmentation scaling (unitary rules). [†] N marks the N at which a rule first becomes effectively ergodic for that boundary convention. Rules reduced by reflection are represented once where noted.

The combinatorial derivation does choose. Writing the chain as “rooms” of length l separated by domain walls, a room contributes $l + 1$ states and consumes $l + 2$ sites, so $D_{\max} = b^N$ with

$$b(l) = (l + 1)^{1/(l+2)}, \quad (1)$$

whose saddle point $d \ln b / dl = 0$ gives the transcendental condition $\ln x = 1 + 1/x$ with $x = l + 1$, i.e. $l \approx 2.5911$. The two integer neighbours of that optimum are precisely the two candidates above, and $l = 3$ is the larger:

$$b(2) = 3^{1/4} = 1.31607 < b(3) = 4^{1/5} = 1.31951.$$

So the asymptotic base is $4^{1/5}$, and at finite N the largest sector is a *mixture* of rooms of length 2 and 3 — which is exactly why neither pure recurrence is exact over the computed range. This is the one base in the report that comes from theory rather than from the data, and it should stay that way.

A second, purely numerical trap sits on the same rule. The BIC model $\ln y = c + \alpha \ln N + \kappa N$ is the right tool for the growth *class*, but its $\alpha \ln N$ term absorbs part of the growth and makes

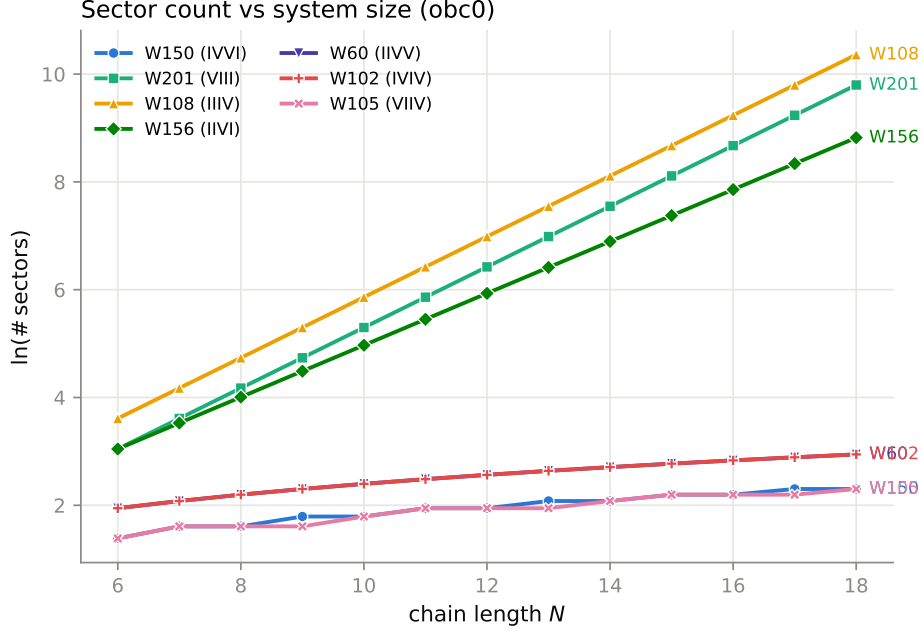


Figure 1: Number of Krylov sectors vs N (obc0), log scale. Exponential fragmentation (156, 108, 201) vs. linear/central-binomial sector counts (60/102, 150/105).

κ a biased rate estimator: it reports base 1.281 for 156/198, far enough below $4^{1/5}$ to look like a different constant. Rates in the growth-base map are therefore taken from the two-parameter fit $\ln y = c + \kappa N$, or from an exact recurrence where one exists.

4 The ergodic rules are ergodic at every N , not just at $N = 6$

The ergodicity verdict as originally produced rested on a single system size. The sweep aborts a unit the moment one component exceeds 90% of 2^N and then stops increasing N for that rule, so an ergodic-flagged unit had exactly one record, at $N = 6$ — and it carried no sector data at all, only the flag. Six sites is far too small to distinguish genuine ergodicity from a finite-size accident, so this claim was re-established from scratch: every ergodic-flagged unit was recomputed with the abort disabled (`run_rule --no-ergodic-abort`), which performs the *full* decomposition and reports how many sectors there really are, over $N = 6 \dots 13$.

The outcome is completely rigid. Across all sixteen units and all eight sizes, only two structures occur — one sector of size 2^N , or one sector of size $2^N - 1$ together with a single frozen basis state — and *each unit gives the same structure at every N* . Nothing drifts, nothing crosses over, and no small sector appears at any size.

rule	tuple	pbc	obc0
51	VVVV	$\{2^N\}$	$\{2^N\}$
57	VIIV	$\{2^N\}$	$\{2^N\}$
99	VVIV	$\{2^N\}$	$\{2^N\}$
54	IVVV	$\{2^N - 1, 1\}$	$\{2^N - 1, 1\}$
60	IIVV	$\{2^N - 1, 1\}$	fragmented ($N + 1$ sectors)
102	IVIV	$\{2^N - 1, 1\}$	fragmented ($N + 1$ sectors)
147	VVVI	$\{2^N - 1, 1\}$	$\{2^N\}$
153	VIVI	$\{2^N - 1, 1\}$	$\{2^N\}$
195	VVII	$\{2^N - 1, 1\}$	$\{2^N\}$

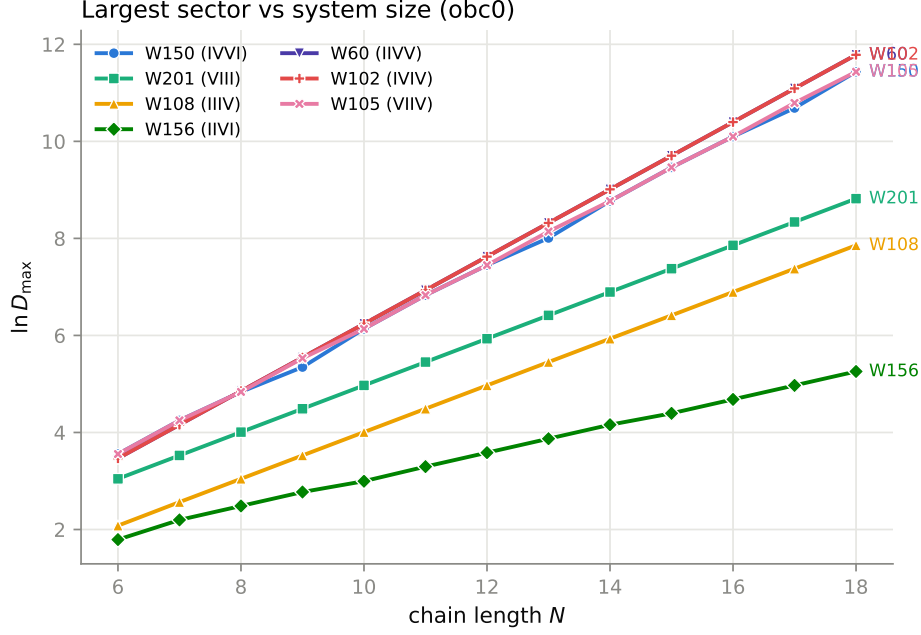


Figure 2: Largest sector $D_{\max}$ vs N (obc0), log scale. Slopes give the exponential base: φ (golden ratio) for 201, $4^{1/5} \approx 1.3195$ for 156/198, 2 for the domain-wall family 60/102.

The pattern is not empirical — it follows from the encoding, which is why more data would add nothing. The all-zeros state has every site sitting in neighbour configuration $(0,0)$, so it is frozen precisely when $r_{00} = \mathbf{I}$; the all-ones state has every site in configuration $(1,1)$ and is frozen precisely when $r_{11} = \mathbf{I}$. Those are the only two configurations that are homogeneous across the whole chain, so no other singleton sector can exist. The table follows immediately: 54/60/102 carry $r_{00} = \mathbf{I}$ and freeze the vacuum under both conventions; 147/153/195 carry $r_{11} = \mathbf{I}$ and freeze $|1 \cdots 1\rangle$ under **pb**c only, because the **obc**0 padding gives the two end sites a 0 neighbour and breaks the homogeneity; 51/57/99 have neither and are a single sector. The singleton was verified directly at $N = 8$ in each case.

The practical upshot for the scaling tables is that these units are not missing data but a trivial growth law: #sectors is constant (1 or 2) and $D_{\max} = 2^N$ or $2^N - 1$, i.e. exponential with base exactly 2.

5 Boundary-sensitive fragmentation

Two rules change fragmentation class with the boundary:

- **Rules 60/102 (IIVV)/(IVIV):** under **obc**0 the domain-wall charge fragments the space into $N+1$ sectors with an extensive dominant sector $D_{\max} = 2^{N-1}$; under **pb**c the periodic seam destroys the charge and the dynamics is ergodic — a single sector of size $2^N - 1$ plus the frozen vacuum $|0 \cdots 0\rangle$ ($D_{\max}/2^N \rightarrow 1$).
- **Rules 150/105:** binomial sector sizes $\binom{N+1}{w}$ under **obc**0 (analysed in §6); they *differ* under **pb**c, where the periodic domain-wall counting is rule-specific.

6 Analytic scaling of the binomial rules 150/105

Rules 150 = (IVVI) and 105 = (VIIV) are the two additive (XOR) elementary rules, $x_i \mapsto x_{i-1} \oplus x_i \oplus x_{i+1}$ and its complement. Under **obc**0 their sectors are not merely “central-binomial-like”:

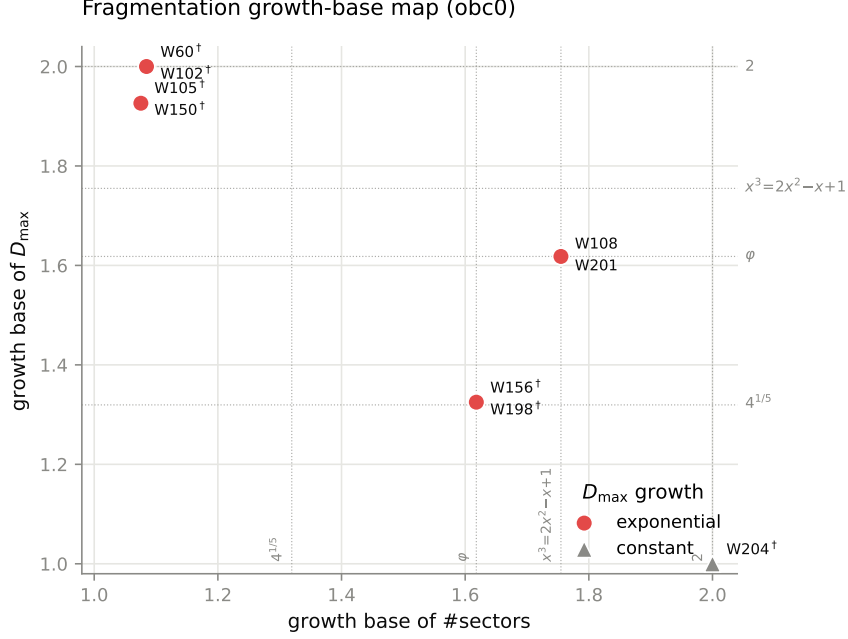


Figure 3: Growth-base map (obc0): the base $b = e^\kappa$ of the sector count against that of the largest sector, one point per rule. Plotting $b \in [1, 2]$ rather than $\kappa \in [0, \ln 2]$ puts the named constants at recognisable positions; the dotted lines mark them. † marks a base from the two-parameter fit $\ln y = c + \kappa N$ rather than an exact recurrence — note in particular that the M2 model $c + \alpha \ln N + \kappa N$ used for the growth *class* is biased low as a rate estimator, reporting 1.281 for 156/198 against its true asymptote $4^{1/5}$.

every sector size is *exactly* a binomial coefficient. Numerically, for all computed $N = 6 \dots 19$ the recurrent-class sizes are

$$\left\{ \binom{N+1}{w} \right\}, \quad \sum_{\text{sectors}} \binom{N+1}{w} = 2^N, \quad (2)$$

i.e. the 2^N states split into fixed-charge Krylov subspaces whose dimensions are the lower half of Pascal's row $N + 1$ (the sum of that half is exactly 2^N). The largest sector is therefore the central binomial,

$$D_{\max} = \binom{N+1}{\lfloor (N+1)/2 \rfloor}. \quad (3)$$

When the two rules agree. For *even* N (so $N + 1$ odd) the row has no self-conjugate entry and 150 and 105 give bit-for-bit identical sector multisets. For *odd* N the central weight $(N+1)/2$ is self-conjugate, and the two rules resolve that one sector differently: at $N \equiv 1 \pmod{4}$, where $(N+1)/2$ is itself odd, 105 keeps the central binomial whole ($D_{\max} = \binom{N+1}{(N+1)/2}$) while 150 splits it into the conjugate pair $\binom{N+1}{(N+1)/2}$ (e.g. $N = 9$: 105 has 252, 150 has 210 + 210). This does not change the exponents below, since $\binom{N+1}{(N-1)/2} / \binom{N+1}{(N+1)/2} = \frac{(N+1)/2}{(N+3)/2} \rightarrow 1$.

Expected exponents. Applying Stirling to (3), $\binom{2m}{m} \sim 4^m / \sqrt{\pi m}$ with $m = \lfloor (N+1)/2 \rfloor$ gives

$$D_{\max} \sim 2^N \sqrt{\frac{8}{\pi(N+1)}}, \quad \ln D_{\max} = N \ln 2 - \frac{1}{2} \ln(N+1) + \frac{1}{2} \ln \frac{8}{\pi}. \quad (4)$$

So the largest sector is *extensive with base exactly 2*, dressed by a $1/\sqrt{N}$ prefactor. In the report’s fit model $\ln y = c + \alpha \ln N + \kappa N$ the prediction is sharp:

$$\kappa = \ln 2 = 0.6931 \text{ (base 2)}, \quad \alpha = -\frac{1}{2}, \quad c = \frac{1}{2} \ln \frac{8}{\pi} = 0.467. \quad (5)$$

The number of sectors is the number of admissible charges, $\sim N/2$ — exactly $\lceil (N+2)/2 \rceil$ for 150, and the same up to a small mod 4 wobble for 105. This is a *linear* law ($\kappa = 0$, base 1, power $\alpha \simeq 1$): subexponentially many sectors each of extensive dimension — the signature of a single $U(1)$ -like domain-wall charge rather than a shattered spectrum.

Which fit to trust, and the mirror of rule 156. The $-\frac{1}{2} \ln N$ term in (4) is not noise: it is the genuine asymptotic form, so the three-parameter M2 model is the correct estimator here and a bare two-parameter fit $\ln y = c + \kappa N$ is *biased low*. On $N = 6 \dots 19$ the bare fit returns base 1.930 — and it creeps upward with the window (1.904, 1.917, 1.923, 1.930 for $N \leq 10, 13, 16, 19$), converging to 2 only as $N \rightarrow \infty$, exactly as the $-\frac{1}{2} \ln N/N$ drag predicts. This is the precise *opposite* of rule 156 (§3), where the true law is a pure geometric b^N and it is the M2 $\alpha \ln N$ term that spuriously eats the growth. The moral is that no single fit is universally right: a domain-wall (binomial) mechanism carries a real half-integer α and must be read from M2, whereas a room-packing (transfer-matrix) mechanism is pure-geometric and must be read from the two-parameter fit. Knowing the mechanism, not the residual, selects the estimator.

Two honest finite-size caveats. The fitted α comes out in the range -0.3 to -0.4 (M2 gives -0.31 ; a fit of $\ln D_{\max} - N \ln 2$ against $\ln N$ gives -0.41) rather than $-\frac{1}{2}$ over this window, because the binomial argument is $N+1$ (not N) and the next Stirling correction $\binom{2m}{m} \propto (1 - \frac{1}{8m})$ adds an $O(1/N)$ term that a three-parameter fit partly absorbs into α ; the residual against the *exact* form (4) confirms $\alpha = -\frac{1}{2}$. And κ from M2 reads 0.684 against the exact $\ln 2 = 0.693$ for the same reason — close, and provably converging, but the exact value comes from (3), not from the regression.

7 Broken inversion symmetry (201 vs. 108)

Rules 201 = (VIII) and 108 = (IIIIV) are $0 \leftrightarrow 1$ spin-flip partners. Because the flip conjugates the Hadamard by X and $XHX \neq \pm H$, the symmetry is broken for the full dynamics. At the level of the sector partition it survives under **pb**c (bit-flip is a graph symmetry: $|XHX| = |H|$) but is broken under **obc**0 by the vacuum-0 padding: there 201 has Fibonacci $D_{\max}$ and 108 shifted-Fibonacci $D_{\max}$ (matching HSF rules 1 and 8). See Report R1 for the exact statement.

8 Reproducibility

All numbers derive from `results/*.jsonl` (engine_version 1a.1); the table and figures are regenerated by `python -m qca_fragmentation.scaling.summary` and `...scaling.figure`. The sweep is idempotent and checkpointed (`checkpoints/manifest.jsonl`).